

Preliminary Examination Paper 1 Solution

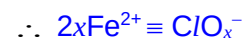
- 1 An acidified solution of the salt KClO_x will oxidise $\text{Fe}^{2+}(\text{aq})$ to $\text{Fe}^{3+}(\text{aq})$ quantitatively, the chlorine being reduced to $\text{Cl}^-(\text{aq})$.

When 0.302 g of the salt KClO_x was reacted with $0.540 \text{ mol dm}^{-3} \text{ Fe}^{2+}(\text{aq})$ in the presence of $\text{H}^+(\text{aq})$, 21.0 cm^3 of $\text{Fe}^{2+}(\text{aq})$ was needed for complete reaction.

Which of the following is the value of x ?

- A 1
B 2
C 3
D 4

Answer: B



OR mole ratio of $\text{ClO}_x^- : \text{Fe}^{2+}$ is

$$1 : 2x$$

$$\text{Amount of Fe}^{2+} \text{ used} = \frac{21.0}{1000} \times 0.540 = 0.01134 \text{ mol}$$

$$\text{Amount of ClO}_x^- = \frac{0.01134}{2x} \text{ mol}$$

$$\text{Amount of KClO}_x = \frac{0.302}{39.1 + 35.5 + x(16)}$$

$$\text{Amount of ClO}_x^- = \frac{0.302}{39.1 + 35.5 + x(16)} = \frac{0.01134}{2x}$$

$$\frac{2(0.302)x}{0.01134} = 74.6 + x(16)$$

$$53.263x - 16x = 74.6$$

$$x = 1.99 \approx \underline{2}$$

- 2 A student made up a 0.10 mol dm^{-3} solution of $\text{Ba(OH)}_2 \cdot 8\text{H}_2\text{O}$ which she found in the laboratory cupboard and left the solution in an open beaker. A week later, she returned to the laboratory, used the solution for titration with $0.10 \text{ mol dm}^{-3} \text{ HCl}$ and was surprised to discover her titres were lower than expected.

Which of the following explains why the values were so low?

- A Some of the barium hydroxide has reacted with carbon dioxide in the air to form solid barium carbonate.
- B Some water had evaporated from the barium hydroxide solution.
- C The concentration of HCl was less than the stated 0.10 mol dm^{-3} .
- D The crystals had less water of crystallisation than stated.

Answer : A

A: will lower the concentration of Ba(OH)_2 and thus a lower than expected titre value.

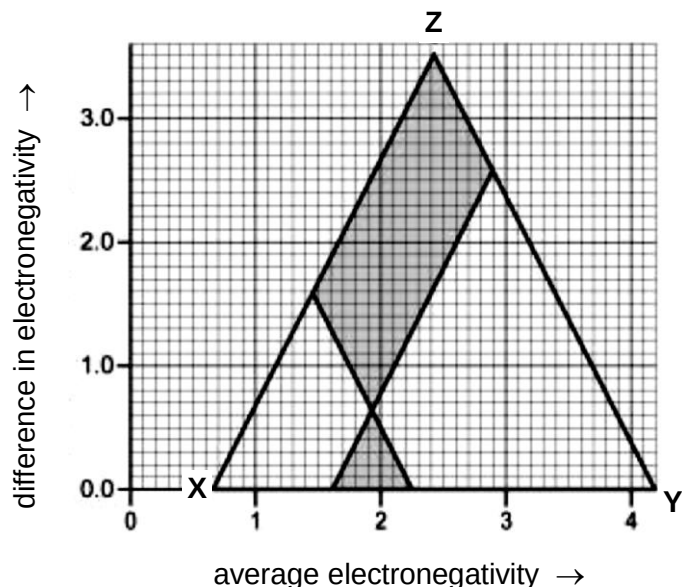
B: will increase the concentration of Ba(OH)_2 and cause the titre value to be higher than expected.

C: will cause the titre value to be higher than expected.

D: will increase the concentration of Ba(OH)_2 and thus a higher than expected titre value.

- 3 The type of bonding between two elements can be rationalised and even predicted using a van Arkel triangle. The triangle is based on electronegativity values. Difference in electronegativity is plotted along the y-axis and average electronegativity is plotted along the x-axis.

What is the type of bonding present at each of these bonding extremes, labelled X, Y and Z on the triangle?



- | | X | Y | Z |
|---|----------|----------|----------|
| A | metallic | covalent | ionic |
| B | covalent | metallic | ionic |
| C | covalent | ionic | metallic |
| D | ionic | covalent | metallic |

Answer: A

Difference in electronegativity is zero for X & Y

⇒ metallic bonding or covalent bonding.

Since metals have low electronegativity

⇒ X is metallic bonding.

- 4 Equimolar amounts of aqueous ClO_2 and hydroxide ions react according to the

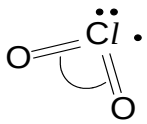
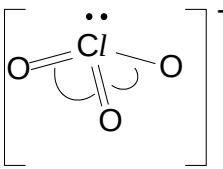
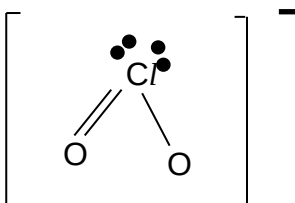
equation below.



Which statement is correct?

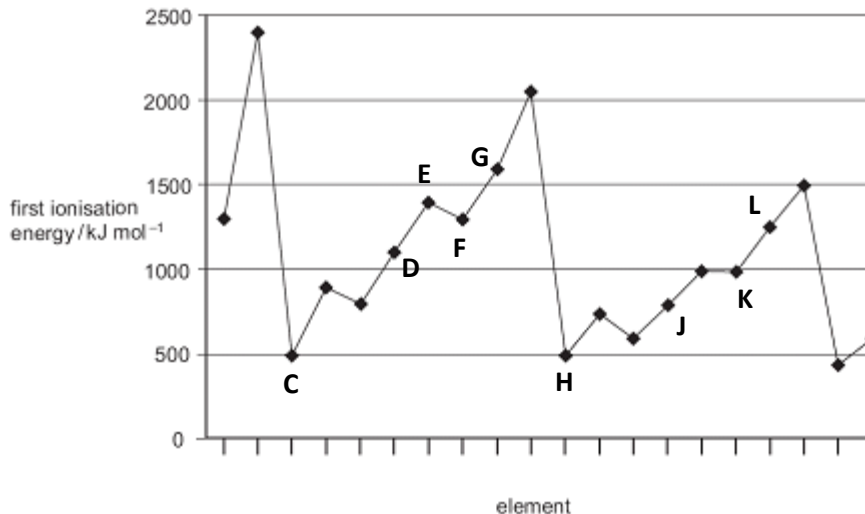
- A** The shape changes from linear in ClO_2 to bent in ClO_2^- .
- B** The shape changes from bent in ClO_2 to trigonal planar in ClO_3^- .
- C** The bond angle changes from about 120° in ClO_2 to 107° in ClO_3^- .
- D** The bond angle changes from about 105° in ClO_2 to 107° in ClO_3^- .

Answer: D

ClO_2  Bond angle $\approx 105^\circ$ Shape: bent	ClO_3^-  Bond angle $\approx 107^\circ$ Shape: trigonal pyramidal	ClO_2^-  Bond angle $\approx 105^\circ$ Shape: bent
---	---	---

5 Use of the Data Booklet is relevant to this question.

The first ionisation energies of successive elements in the Periodic Table are represented in the graph.



Which of the following statements about elements **C** to **L** are correct?

- 1 **E** and **J** have half-filled p-orbitals.
- 2 Chlorides of **D** and **J** undergo hydrolysis.
- 3 **F** and **K** reacts to form acidic compounds.
- 4 **D** and **F** forms compounds with formula **DF** and **DF₂**.
- 5 **G** and **L** forms compounds with variable oxidation states from -1 to $+7$.

- A** 1 and 4 only
- B** 2 and 5 only
- C** 1, 3 and 4 only
- D** 2, 3 and 5 only

Answer : C (Statements 1, 3, 4 only)

Elements **C** and **H** are from group 1 as they have the lowest first I.E.
They are lithium and sodium respectively.

1 is correct.

E is from Group 15 and **J** is from Group 14.

Element **E** is nitrogen and has half-filled 2p orbitals.

Element **J** is silicon, and Si^- has half-filled 3p orbitals.

2 is wrong.

D and **J** are from Group 14.

D is a Period 2 element and is carbon. It cannot undergo hydrolysis as they do not have low lying d-orbitals to accommodate the lone pair of electrons from water, hence no hydrolysis can occur.

3 is correct.

Element **F** is oxygen and element **K** is sulfur. SO_2 and SO_3 are acidic in nature.

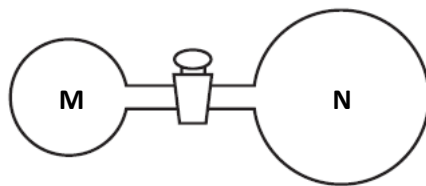
4 is correct.

Element **D** is carbon and **F** is oxygen. Hence **DF** and **DF₂** is possible (ie, CO and CO_2).

5 is wrong.

Elements **G** is fluorine and **L** is chlorine. Chlorine can have variable oxidation states from -1 to $+7$, but fluorine only exists in states of -1 and 0 .

- 6** Two glass vessels **M** and **N** are connected by a closed valve.



M contains helium at 25°C at a pressure of $1 \times 10^5 \text{ Pa}$. **N** has been evacuated, and has three times the volume of **M**. In an experiment, the valve is opened and the whole set-up placed in boiling water.

What is the final pressure in the system?

A $3.13 \times 10^4 \text{ Pa}$

B $3.76 \times 10^4 \text{ Pa}$

C $1.00 \times 10^5 \text{ Pa}$

D $1.33 \times 10^5 \text{ Pa}$

Answer: A

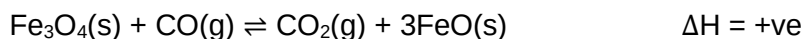
Let the volume of M be v , hence volume of N is $3v$. Total volume = $4v$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{10^5 \times v}{298} = \frac{P_2 \times 4v}{373}$$

$$P_2 = 3.13 \times 10^4 \text{ Pa}$$

7 Consider the following equilibrium system:



Given that $K_p = K_c (\text{RT})^{\Delta n}$, where Δn is the sum of gaseous products – sum of gaseous reactants.

Which of the following statements are correct?

- 1 $K_c = K_p$ for this system.
- 2 The equilibrium constant, K_c , for this reaction has no units.
- 3 Adding FeO causes the position of equilibrium to shift to the left.
- 4 Increasing the temperature causes the position of equilibrium to shift to the right.

- A** 1 and 2
B 1 and 3
C 2, 3 and 4
D 1, 2 and 4

Answer : **D**

1 is correct: $\Delta n = 0$, $K_c = K_p$ for this system

2 is correct: $K_c = \frac{[\text{CO}_2]}{[\text{CO}]}$ no units

3 is incorrect: FeO is a solid. Position of equilibrium is not affected by it.

4 is correct: Increasing temperature would favour the forward endothermic reaction. Position of equilibrium would shift to the right.

8 In an experiment to measure the enthalpy change of neutralisation, 20 cm³ of aqueous sulfuric acid containing 0.02 mol of H₂SO₄ is placed in a plastic cup of negligible heat capacity.

A 20 cm³ sample of aqueous sodium hydroxide containing 0.04 mol of NaOH, at the same initial temperature, is added and the temperature rises by 15 K.

If the heat capacity per unit volume of the final solution is 4.2 J K⁻¹ cm⁻³, what is the enthalpy change of neutralisation?

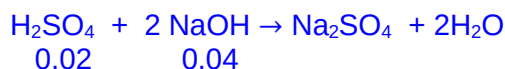
A $-\left(\frac{20 \times 4.2 \times 15}{0.04 \times 1000}\right) \text{ kJ mol}^{-1}$

B $-\left(\frac{40 \times 4.2 \times 15}{0.04 \times 1000}\right) \text{ kJ mol}^{-1}$

C $-\left(\frac{20 \times 4.2 \times 15}{0.02 \times 1000}\right) \text{ kJ mol}^{-1}$

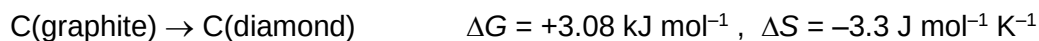
D $-\left(\frac{40 \times 4.2 \times 15}{0.02 \times 1000}\right) \text{ kJ mol}^{-1}$

Answer : **B**



$$Q = -\frac{mc\Delta T}{n_{\text{H}_2\text{O}}} = -\frac{40 \times 4.2 \times 15}{0.04 \times 1000} = -63 \text{ kJ mol}^{-1}$$

- 9 The conversion of graphite into diamond is a non-spontaneous reaction at 298 K. The entropy change of this reaction is negative at 298 K.



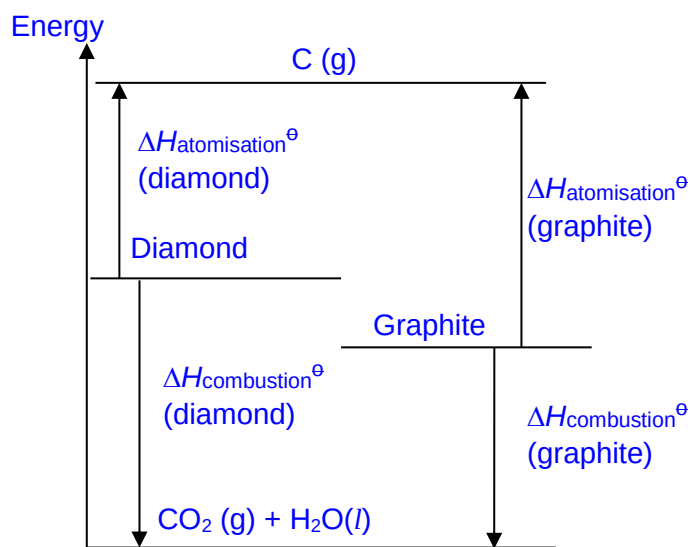
With reference to the enthalpy change, ΔH , of the reaction above, which statements are correct?

- 1 Enthalpy change of atomisation of diamond is more endothermic than that of graphite.
- 2 Enthalpy change of combustion of diamond is more exothermic than that of graphite.
- 3 Bond energy of carbon-carbon bonds in graphite is greater than that in diamond.
- 4 The activation energy to convert graphite to diamond is lower than the activation energy for the reverse reaction.

- A 1 and 4
B 2 and 3
C 1, 2 and 3
D 2, 3 and 4

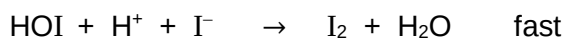
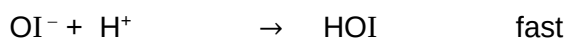
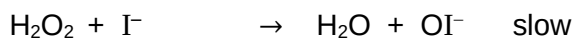
Answer : **B (2 & 3 are correct)**

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ 3.08 &= \Delta H - (298) (-3.3/1000) \\ \Delta H &= + 2.10 \text{ kJ mol}^{-1}\end{aligned}$$

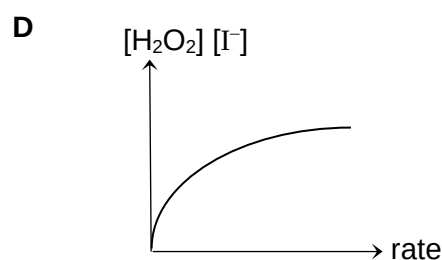
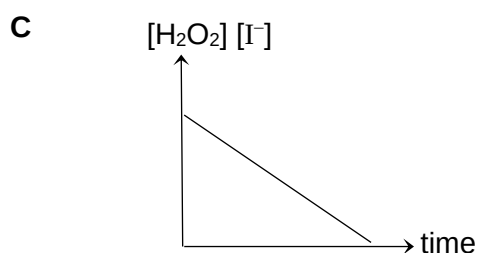
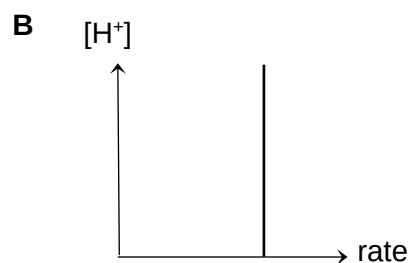
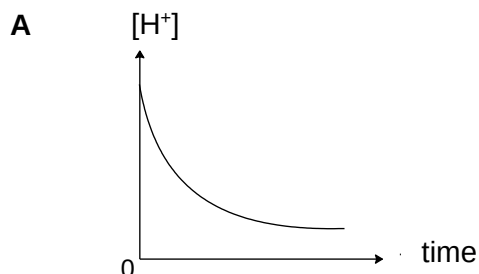


The carbon in graphite is sp^2 hybridised and carbon in diamond is sp^3 hybridised. Since the s character is higher in C-C bond in graphite, it will be a stronger bond than C-C bond in diamond.

- 10 The reaction between acidified KI and aqueous H_2O_2 involves the following steps:



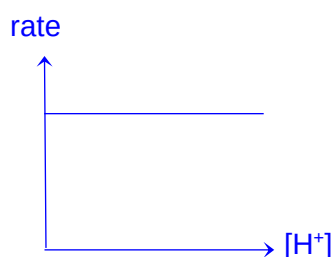
Which graph would be obtained?



Answer: **B**

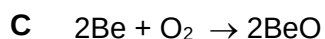
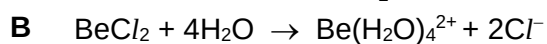
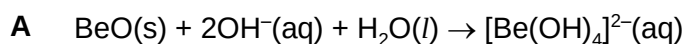
Base on the slow step, $\text{rate} = k [\text{H}_2\text{O}_2] [\text{I}^-]$

The rate is independent of $[\text{H}^+]$.



- 11 Compounds of beryllium are amphoteric, like compounds of aluminium.

In which equations is the beryllium-containing reactant acting as a Brønsted-Lowry acid?



Answer: **D**

A Brønsted-Lowry acid is a proton donor.

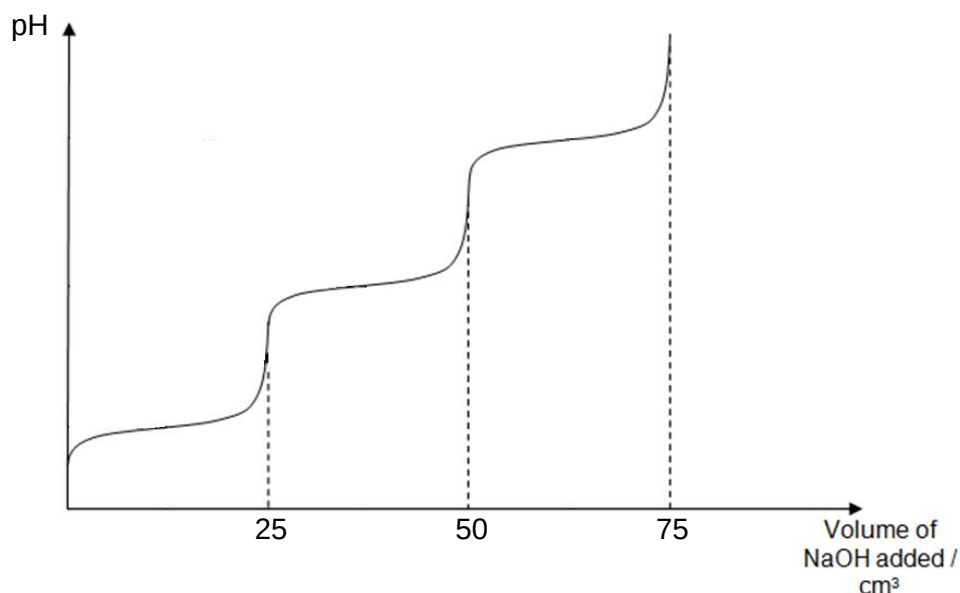
A: This is neutralisation reaction but BeO is not a Brønsted-Lowry acid as it does not donate a H^+ ion.

B: This is simply a dissociation reaction and BeCl_2 is not a Brønsted-Lowry acid as it does not donate a H^+ ion.

C: This is a combustion reaction.

D: $\text{Be}(\text{H}_2\text{O})_2(\text{OH})_2$ is acting as a Brønsted-Lowry acid as 2 H_2O molecules are deprotonated to form 2 additional OH^- ions and result in a negatively charged $\text{Be}(\text{OH})_4^{2-}$ ion.

- 12** 25 cm^3 of 1 mol dm^{-3} of the fully protonated form of compound **O** was titrated against a standard sodium hydroxide solution of a similar concentration and the following titration curve was obtained.



Which of the following compounds is a possible identity of compound **O**?

- A** $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CONH}_2)\text{COOH}$
- B** $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CH}_2\text{OH})\text{COOH}$
- C** $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2)\text{COOH}$
- D** $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CHO})\text{COOH}$

Answer: C

All amine groups are protonated and can act as a bronsted-lowry acid.

A : only two acidic groups as the amide on the R group is neutral

B : only two acidic groups as the OH on the R group is not able to be neutralised by NaOH

C : three acidic groups

D : only two acidic groups as the aldehyde on the R group is also neutral

- 13 A sparingly soluble calcium salt dissociates in solution according to the equation:



If the solubility product of Ca_3Q_2 is p , what is the concentration of Q^{3-} at equilibrium in a saturated solution of Ca_3Q_2 (aq)?

- A $2p^{\frac{1}{5}}$
B $2\left(\frac{p}{36}\right)^{\frac{1}{5}}$
C $2\left(\frac{p}{72}\right)^{\frac{1}{5}}$
D $2\left(\frac{p}{108}\right)^{\frac{1}{5}}$

Answer: D



Let solubility be $x \text{ mol dm}^{-3}$

$$K_{\text{sp}} = [\text{Ca}^{2+}]^3 [\text{Q}^{3-}]^2$$

$$p = (3x)^3 (2x)^2$$

$$p = (27x^3)(4x^2)$$

$$p = 108x^5$$

$$x = \left(\frac{p}{108}\right)^{\frac{1}{5}}$$

$$2x = 2\left(\frac{p}{108}\right)^{\frac{1}{5}}$$

- 14 A transition metal cation can exist in a 'high spin' or 'low spin' state while exposed to different ligands in an octahedral complex.

A 'high-spin' state has the electrons occupying all the d-orbitals singly first, before pairing in the lower energy d-orbitals.

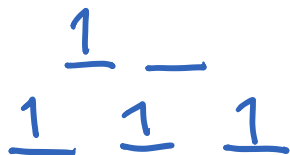
A 'low-spin' state has the lower energy d-orbitals filled first, pairing up if necessary, before the higher energy d-orbitals are filled.

Which cation would have the largest number of unpaired electrons?

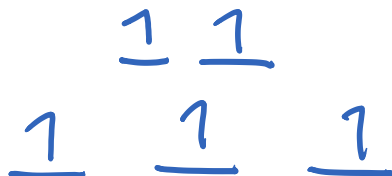
- A 'High-spin' Mn^{3+}
B 'High-spin' Fe^{3+}
C 'Low-spin' Co^{2+}
D 'Low-spin' Ni^{4+}

Answer: B

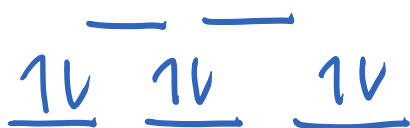
A: 'High-spin' $\text{Mn}^{3+} \rightarrow 4$ unpaired electrons



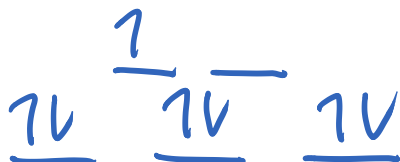
B: 'High-spin' $\text{Fe}^{3+} \rightarrow 5$ unpaired electrons



D: 'Low-spin' $\text{Ni}^{4+} \rightarrow$ no unpaired electrons



C: 'Low-spin' $\text{Co}^{2+} \rightarrow 1$ unpaired electron



15 Which of the following is **not** a result of the small energy gap between 3d and 4s orbitals?

- A Transition metals have higher electrical conductivity than aluminium.
- B Transition metals can exhibit variable oxidation states.
- C Transition metals have higher melting point than calcium.
- D Transition metals ions are usually coloured.

Answer : D

The smaller energy gap allow for valence electrons to be removed from both 3d and 4s orbitals. This leads to transition metals having higher conductivity, having variable oxidations states and higher melting point.

The energy gap is due to absorption in the visible light region due to d-d orbital splitting and transition.

16 Use of the Data Booklet is relevant to this question.

Two catalysed reactions and their respective catalysts are shown below.

Reaction	Equation	Catalyst
I	$\text{S}_2\text{O}_8^{2-} + 2\text{I}^- \rightarrow 2\text{SO}_4^{2-} + \text{I}_2$	Fe^{3+}
II	$2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$	Mn^{2+}

Which statements about the reactions and the catalysts are correct?

- 1** Mn^{2+} is an auto-catalyst in reaction **II**.
- 2** Fe^{2+} could replace Fe^{3+} as a heterogeneous catalyst in reaction **I**.
- 3** Mn^{2+} could replace Fe^{3+} as a homogeneous catalyst in reaction **I**.
- 4** Both catalysts increase the energetic feasibility of their respective reactions.

- A** 1 and 3
B 2 and 3
C 1, 2 and 4 only
D 2, 3 and 4 only
 Answer: A (1 and 3 only)

1 is correct: Mn^{2+} is a catalyst as well as a product of reaction **II**. Thus, Mn^{2+} is an auto-catalyst.

2 is incorrect: Fe^{2+} can replace Fe^{3+} as homogeneous catalyst, not heterogeneous catalyst.

3 is correct:

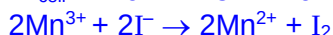
From Data Booklet: $E^\circ(\text{S}_2\text{O}_8^{2-}/\text{SO}_4^{2-}) = +2.01 \text{ V}$

$E^\circ(\text{I}_2/\text{I}^-) = +0.54 \text{ V}$

$E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+}) = +1.54 \text{ V}$



$$E^\circ_{\text{cell}} = 2.01 - 1.54 = +0.47 \text{ V} > 0$$



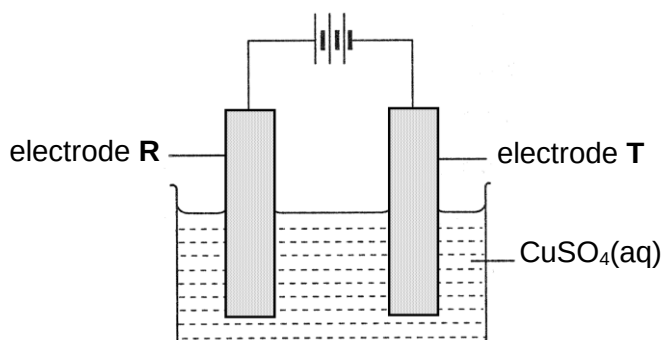
$$E^\circ_{\text{cell}} = 1.54 - 0.54 = +1.00 \text{ V} > 0$$

Thus, Mn^{2+} could replace Fe^{3+} as a homogeneous catalyst in reaction **I**.

4 is incorrect: ΔH remains the same for a catalysed or uncatalysed reaction. Reaction will remain just as energetically feasible.

17 Use of the Data Booklet is relevant to this question.

Impure copper obtained from copper ores can be purified by electrolysis as shown below. The cell potential is adjusted such that copper of the impure copper electrode dissolves and impurities such as silver, iron and zinc can be removed through this process.



Which statement is **not** correct regarding the above electrolytic process?

- A Reduction occurs at electrode T.
- B Electrode R contains impure copper.
- C Zinc impurity goes into the solution as Zn^{2+} ions.
- D Iron and silver impurities fall to the bottom as sludge.

Answer: D

Option A is correct: Electrode T is the cathode where reduction occurs.

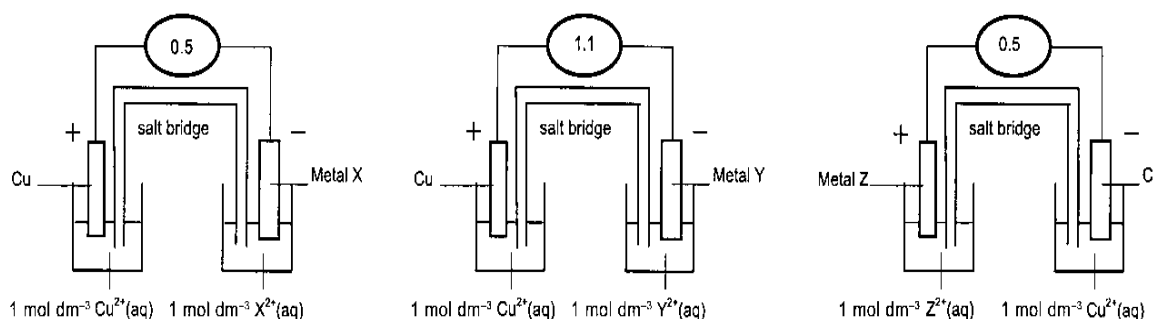
Option B is correct. Electrode R is the anode which contains impure copper.

Option C is correct. $E^\ominus(\text{Zn}^{2+}/\text{Zn})$ is more negative than $E^\ominus(\text{Cu}^{2+}/\text{Cu})$, hence Zn will be oxidised to form Zn^{2+} ions.

Option D is incorrect. $E^\ominus(\text{Fe}^{2+}/\text{Fe})$ is more negative than $E^\ominus(\text{Cu}^{2+}/\text{Cu})$, hence Fe will be oxidised to form Fe^{2+} ions, instead of falling to the bottom as sludge.

18 Use of the Data Booklet is relevant to this question.

Three electrochemical cells are set up as shown below. The *e.m.f* in volts is shown on each voltmeter.



These *e.m.f* indicate the order of reactivity of the metals. The order of the weakest to the strongest reducing agent is

- A X, Y, Cu, Z
- B Cu, Z, X, Y
- C Z, Cu, X, Y
- D Y, X, Z, Cu

Answer: C

$$E^\ominus(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$$

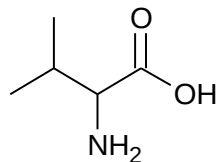
For electrochemical cell on the left, Cu^{2+}/Cu half-cell is the cathode. Hence, $E^\ominus(\text{X}^{2+}/\text{X}) = 0.34 - 0.5 = -0.16 \text{ V}$

For electrochemical cell in the centre, Cu^{2+}/Cu half-cell is the cathode. Hence, $E^\ominus(\text{Y}^{2+}/\text{Y}) = 0.34 - 1.1 = -0.76 \text{ V}$

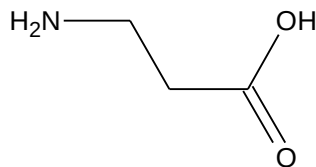
For electrochemical cell on the right, Cu^{2+}/Cu half-cell is the anode. Hence, $E^\ominus(\text{Z}^{2+}/\text{Z}) = 0.5 + 0.34 = +0.84 \text{ V}$

Order of weakest reducing reagent (least easily oxidised, least negative $E^\ominus$ value) to strongest reducing agent (most easily oxidised, most negative $E^\ominus$ value): Z, Cu, X, Y

19 Valine is a naturally occurring amino acid.



Valine



3-aminopropanoic acid

Which statements about valine and 3-aminopropanoic acid are correct?

- A Both compounds are able to form zwitterions.
- B Both compounds are able to react with ethanoic acid to give amide.
- C Both compounds are able to react with NaBH_4 in methanol to give alcohol.
- D Both compounds are soluble in water due to the formation of intermolecular hydrogen bonding.

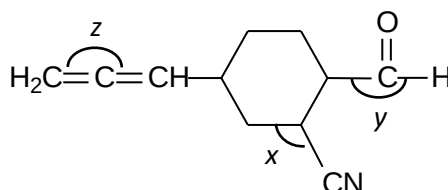
Answer: A

Option B is incorrect: Both compounds are not able to react with ethanoic acid to give amide. Acid-base neutralization would occur, producing a salt.

Option C is incorrect: Both compounds are not able to react with NaBH_4 in methanol to give alcohol. They must react with LiAlH_4 in dry ether.

Option D is incorrect: Both valine and 3-aminopropanoic acid are soluble in water due to formation of ion-dipole interactions, not hydrogen bonding.

20 Which of the following statements are correct about compound **U**?



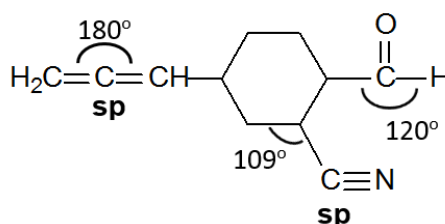
compound **U**

- 1 The bond angles in compound **U** increase in the order $x < y < z$.
- 2 There is only one sp hybridised carbon atom in compound **U**.
- 3 There are 26 σ and 5 π bonds.
- 4 Compound **U** is planar.

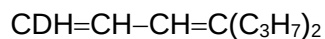
- A 1 and 2
- B 1 and 3
- C 1, 2 and 4 only
- D 2, 3 and 4 only

Answer : **B** (1 and 3 are correct)

26 σ and 5 π bonds.



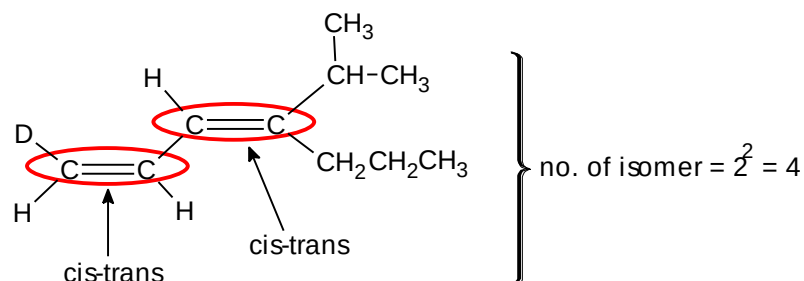
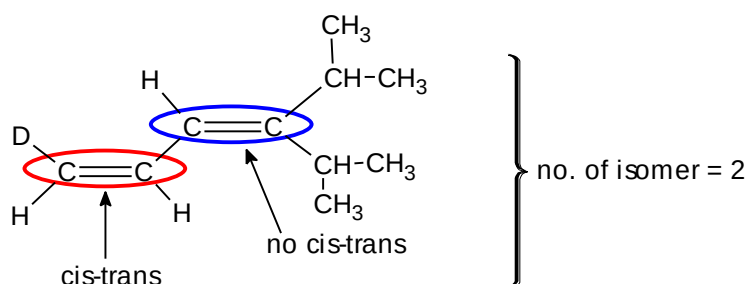
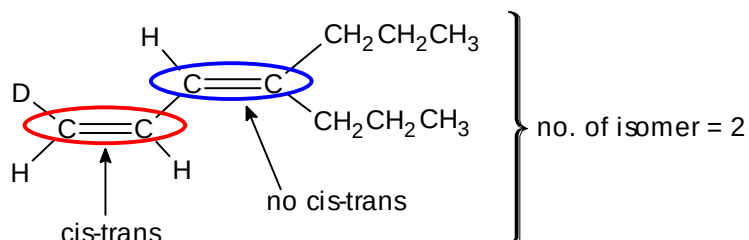
- 21 Deuterium (D or ^2H) is a heavy isotope of hydrogen. A deuterio-hydrocarbon has the part structural formula shown below.



What is the total number of isomers with the above part-structural formula?

- A 2
B 4
C 8
D 10

Answer: C



Total no. of isomers = 8

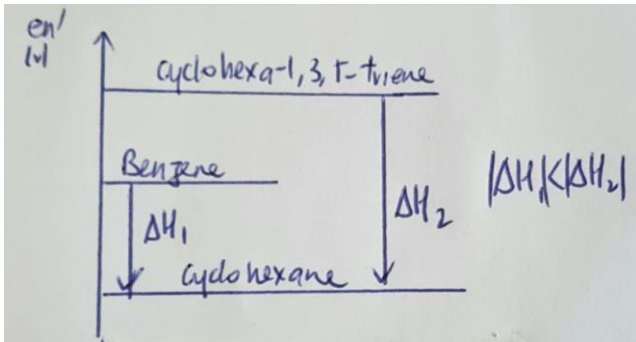
- 22 The molecule of benzene, C_6H_6 is a regular hexagon in which the π electrons are described as delocalised.

Which of the following statements support this structure?

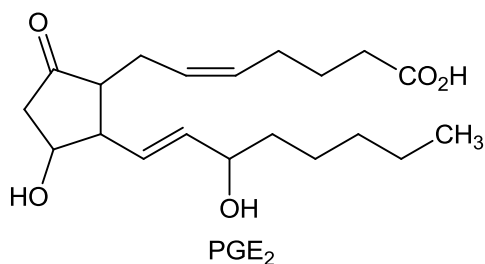
- 1 Benzene is a good conductor of electricity.
- 2 Addition reactions of benzene take place more easily than substitution.
- 3 All C–C bonds lengths in benzene are intermediate between C–C bond in an alkane and C=C bond in an alkene.
- 4 The enthalpy change of hydrogenation of benzene is less exothermic than that predicted for cyclohexatriene.

- A 1 and 3
B 3 and 4
C 1, 2 and 3 only
D 1, 2 and 4 only

Answer : B (3 and 4 are correct)

1	Benzene is a simple molecular structure and does not conduct electricity.
2	Benzene will undergo substitution reactions easily as a result of its resonance stability.
3	The bond order of all C–C bonds in benzene are 1.5 due to resonance which arises from the continuous overlap of unhybridised p-orbitals.
4	Benzene is more stable than cyclohexa-1,3,5-triene due to resonance. Thus, it exists at a lower energy level (see below). Hydrogenation of benzene gives cyclohexane. 

23 PGE₂ is a prostaglandin that has pharmacological activity.



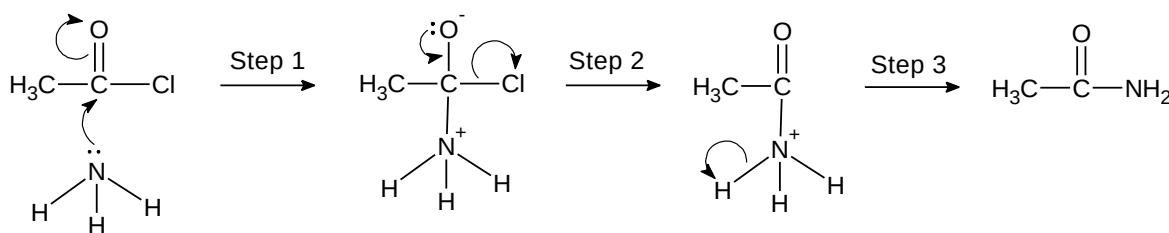
Which of the following is incorrect?

	Reducing agent	Number of hydrogen atoms incorporated per molecule of PGE ₂
A	H ₂ / Ni	6
B	Na in ethanol	8
C	NaBH ₄ in methanol	6
D	LiAlH ₄ in dry ether	4

Answer : C

- A** H₂ / Ni will reduce the ketone and the two C=C. H₂ / Ni cannot reduce the carboxylic acid. Hence **6 hydrogen atoms** will be incorporated in the product.
- B** Na in ethanol will reduce all the functional groups – ketone, the two C=C. H₂ / Ni and the carboxylic acid. Hence **8 hydrogen atoms** will be incorporated in the product.
- C** NaBH₄ in methanol will only reduce the ketone. Carboxylic acid and C=C cannot be reduced. Hence **only 2 hydrogen atoms** will be incorporated in the product.
- D** LiAlH₄ in dry ether will reduce the ketone and carboxylic acid. C=C cannot be reduced. Hence **4 hydrogen atoms** will be incorporated in the product.

24 A mechanism for the reaction between ammonia and ethanoyl chloride is given below.



Which of the following statements are correct?

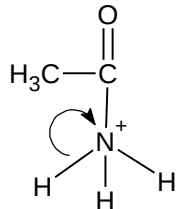
- 1 The ammonia behaves as a nucleophile.
- 2 The loss of proton in **step 3** is shown incorrectly.
- 3 The ammonia attacks an electron-deficient carbon atom.
- 4 The rate of reaction will increase if ethanoyl bromide is used.

- A 1, 2, 3 and 4.
B 1, 3 and 4 only.
C 1 and 3 only.
D 2 and 4 only.

Answer : A (1,2,3 and 4)

1: Correct. NH_3 has a lone pair of electrons and uses it to attack the electron deficient carbon.

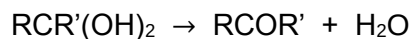
2: Correct. The arrow in step 3 is shown incorrectly. The correct arrow movement should be



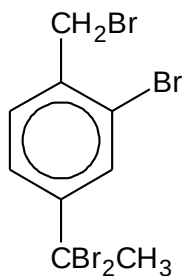
3: Correct. The carbon has two electron withdrawing groups (oxygen and chlorine) bonded to it, resulting in an very electron-deficient carbon.

4: Step 2 involves the breakage of C-Cl bond. Comparing the bond energy of C-Cl (340 kJ) and C-Br (280 kJ), less energy required to break the C-Br bond, activation energy for the reaction involving ethanoyl bromide will be lower, hence the rate of reaction will be faster.

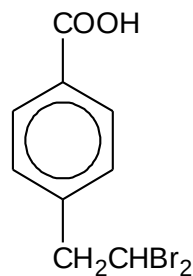
- 25 Diols formed on the same carbon atom are generally unstable and will undergo the following reaction.



Compounds **V** and **W** are both reacted with hot aqueous NaOH followed by dilute HCl at room temperature.



Compound **V**



Compound **W**

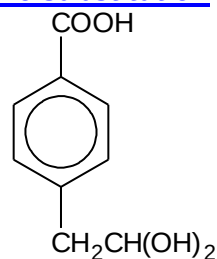
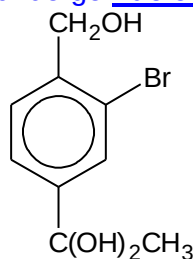
Which reagents could be used to distinguish the final products of **V** and **W**?

- 1 2,4 DNPH
- 2 Fehling's solution
- 3 Sodium carbonate
- 4 Hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$
- 5 Iodine in aqueous NaOH

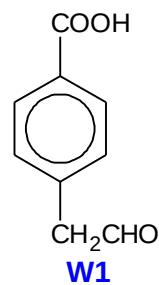
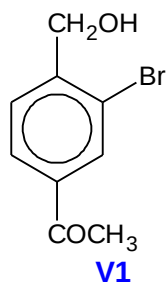
- A 1 and 2 only
- B 3 and 4 only
- C 1, 4 and 5 only
- D 2, 3 and 5 only

Answer : D (2, 3 and 5 only)

The halogenoalkanes will undergo nucleophilic substitution to give alcohols.



Diols on the same C atom are unstable, hence they decompose (with the loss of water) to give the respective carbonyl compounds.



- | | | |
|---|------------|------------|
| 1 | +ve for V1 | +ve for W1 |
| 2 | -ve for V1 | +ve for W1 |
| 3 | -ve for V1 | +ve for W1 |
| 4 | +ve for V1 | +ve for W1 |
| 5 | +ve for V1 | -ve for W1 |

- 26 The ozone depletion potential (ODP) of a chemical compound is the relative amount of degradation to the ozone layer it can cause, with trichlorofluoromethane being fixed at an ODP of 1.0.

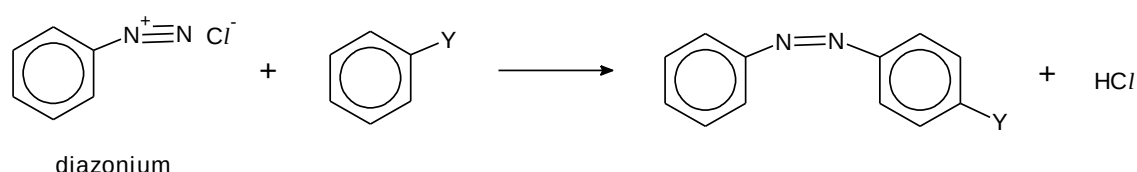
Which of the following compounds has an ODP of greater than 1.0?

- A methane
- B dichlorodifluoromethane
- C chlorotrifluoromethane
- D bromotrifluoromethane

Answer: D

C-Br is weaker than C-Cl bond. Less energy is required to homolytically break C-Br bond, hence the bromide radical will be easier to generate. Bromide radical would then break down ozone.

- 27 Azo coupling is an organic reaction between a diazonium compound and another aromatic compound that produces an azo compound.



Which of the following correctly represents the type of reaction and the role of diazonium compound?

- | | <u>Type of reaction</u> | <u>Role of diazonium</u> |
|---|----------------------------|--------------------------|
| A | nucleophilic substitution | nucleophile |
| B | electrophilic substitution | electrophile |
| C | nucleophilic substitution | electrophile |
| D | electrophilic substitution | nucleophile |

Answer: B

Diazonium has a positive charge, hence it is an electrophile. Electrophilic substitution occurs on the benzene ring.

- 28 What is the order of increasing pH of 1 mol dm⁻³ of each organic compound?

- A CH₃CO₂Na, CH₃CO₂H, CF₃CO₂H, HOCH₂CHO
- B CH₃CO₂Na, HOCH₂CHO, CH₃CO₂H, CF₃CO₂H
- C CF₃CO₂H, CH₃CO₂H, HOCH₂CHO, CH₃CO₂Na
- D CF₃CO₂H, CH₃CO₂H, CH₃CO₂Na, HOCH₂CHO

Answer: C

CF₃CO₂H and CH₃CO₂H are both acids, both pH < 7, CF₃CO₂H has 3 electron-withdrawing fluorine which stabilises the conjugate to greater extent, CF₃CO₂H is the strongest acid, hence lowest pH.

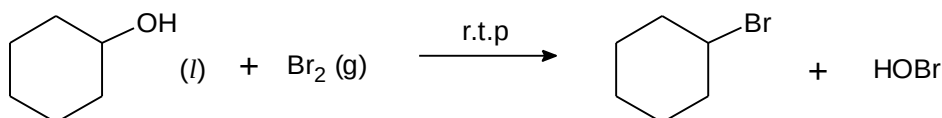
HOCH₂CHO has alcohol and aldehyde hence it is neutral, pH = 7.

CH₃CO₂Na is the conjugate base of CH₃CO₂H, hence it has a pH > 7.

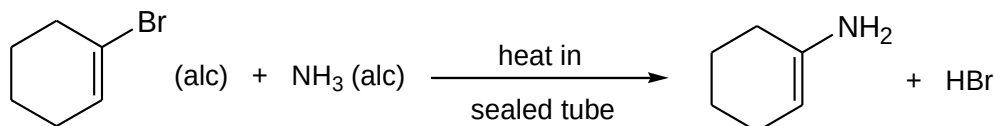
29 Which reaction gives the best yield of products?

[(alc) indicates an alcoholic solution.]

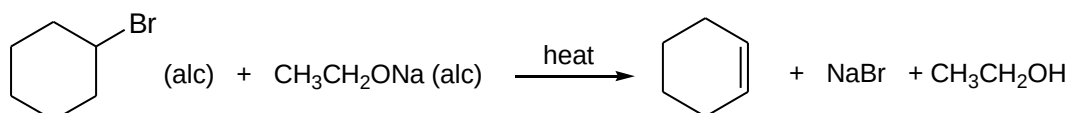
A



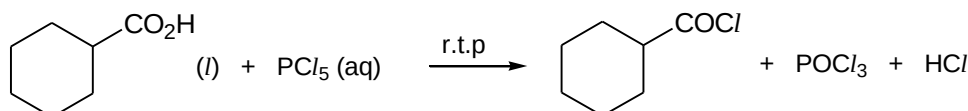
B



C



D



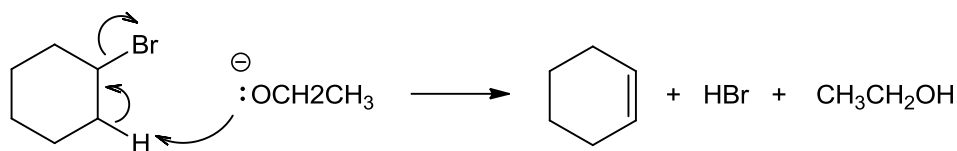
Answer: C

A Substitution of OH with Br requires HBr or PBr₃, as the C-O is strong and difficult to break.

B Br attached to C=C is unreactive towards nucleophilic substitution as the p orbital on Br overlaps with the π orbital of C=C, resulting in partial double bond character for the carbon-bromine bond which requires more energy to break.

C Elimination of HBr can take place in presence of a strong base in alcoholic medium.

CH₃CH₂O⁻ is a stronger base than OH⁻ since the ethyl group is electron donating which intensify the negative charge on oxygen. Thus it is able to abstract the H atom on the carbon next to carbon attached to Br.

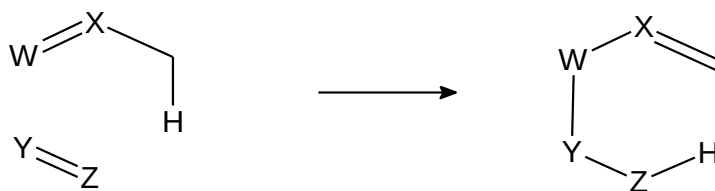


Na⁺ is a spectator ion.

D PCl₅ will hydrolyse in aqueous medium and hence anhydrous PCl₅ should be used instead.



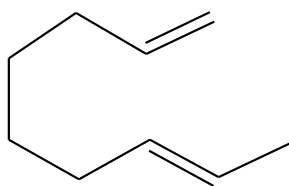
- 30 The Alder-ene reaction is a reaction between two alkenes and is capable of forming new carbon-carbon bonds. An example of the Alder-ene reaction is given below:



Note: W, X, Y & Z are carbon atoms and other hydrogens are not shown

The Alder-ene reaction typically produces many products as the roles of the two alkenes can be exchanged.

Which compound will **not** be formed when the following compound undergoes Alder-ene reaction?

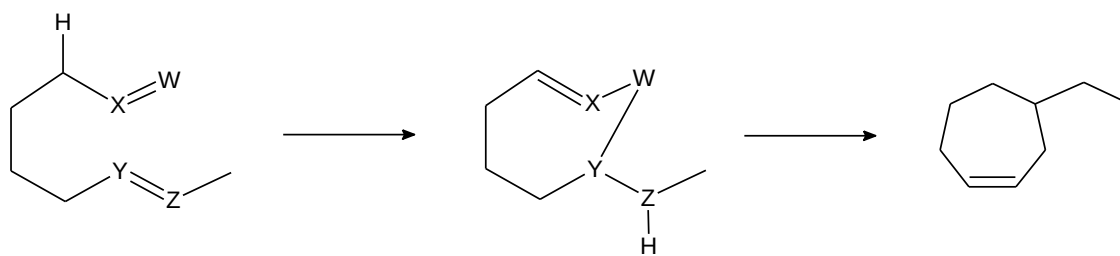


- A
- B
- C
- D

Answer: A



C:



D:

